

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular/Supplementary Winter Examination – 2024

Course: B.Tech

Semester : VII

Branch: Electrical Engineering / Electrical Engineering (Electronics and Power)/Electrical & Electronics Engg. / Electrical & Power Engineering.

Subject Code & Name: BTEEOE705B Electric and Hybrid Electric Vehicles

Max Marks: 60

Date: 21/02/2025

Duration: 3 Hr.

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

(Level/CO) Marks

Q. 1 Objective type questions. (Compulsory Question)

12

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|---|---|-----|---|
| 1 | Which of the following is a primary reason for the development of hybrid and electric vehicles?
a) To reduce vehicle manufacturing costs b) To increase vehicle speed c) To mitigate environmental impact d) To simplify vehicle design | CO1 | 1 |
| 2 | Which parameter is NOT considered in the mathematical modeling of vehicle performance?
a) Aerodynamic drag b) Rolling resistance c) Engine displacement d) Transmission characteristics | CO1 | 1 |
| 3 | Which of the following is NOT a hybrid drive-train topology?
a) Series b) Parallel c) Combined d) Sequential | CO1 | 1 |
| 4 | What is the main purpose of power flow control in hybrid drive-trains?
a) To maximize speed b) To minimize fuel consumption c) To increase battery life d) To simplify vehicle design | CO1 | 1 |
| 5 | What is the primary advantage of hybrid traction systems?
a) Improved vehicle aesthetics b) Increased fuel efficiency c) Reduced maintenance requirements d) Faster acceleration | CO2 | 1 |
| 6 | Electric drive-trains primarily rely on which type of energy source?
a) Fossil fuels b) Batteries c) Nuclear energy d) Flywheels | CO2 | 1 |
| 7 | Which drive system is known for its simple construction and high reliability?
a) DC motor drive b) Induction motor drive c) Switch reluctance motor drive d) Permanent magnet motor drive | CO2 | 1 |
| 8 | Which of the following energy storage systems is most used in hybrid electric vehicles?
a) Supercapacitors b) Flywheels c) Lithium-ion batteries d) Fuel cells | CO3 | 1 |

9	Fuel cells primarily use which chemical reaction to produce energy?	CO3	1
	a) Combustion b) Electrolysis c) Oxidation-reduction d) Fission		
10	What is the key challenge in hybridizing different energy storage devices?	CO3	1
	a) Cost b) Compatibility c) Weight d) Power density		
11	What is the purpose of matching the electric machine and the internal combustion engine (ICE)?	CO3	1
	a) To optimize efficiency b) To reduce vehicle weight c) To simplify the powertrain design d) To increase vehicle speed		
12	Which factor influences the selection of power electronics for hybrid vehicles?	CO4	1
	a) Fuel efficiency b) Power rating c) Vehicle aesthetics d) Cost of materials		
Q. 2 Solve the following.			12
A)	Explain the history and evolution of hybrid and electric vehicles. Discuss their social and environmental importance.	CO2	6
B)	Compare and contrast conventional vehicles with hybrid electric vehicles in terms of energy efficiency, emissions, and performance.	CO1	6
Q.3 Solve the following.			12
A)	Explain the basic concept of hybrid traction and discuss the various hybrid drive-train topologies in detail.	CO2	6
B)	Describe the process of power flow control in hybrid drive-train topologies and analyze its impact on fuel efficiency.	CO2	6
Q. 4 Solve Any Two of the following.			12
A)	Explain the configuration and control of DC Motor drives, Induction Motor drives.	CO3	6
B)	Discuss the significance of power converters in electric vehicles and analyze their role in drive system efficiency.	CO3	6
C)	Explain how the drive system efficiency is achieved in hybrid and electric vehicles. Discuss the role of motor control techniques.	CO3	6
Q.5 Solve Any Two of the following.			12
A)	Explain the energy storage requirements in hybrid and electric vehicles.	CO3	6
B)	Discuss the working, advantages, and limitations of fuel cell-based energy storage systems in hybrid and electric vehicles.	CO3	6
C)	Explain the concept of hybridization of energy storage devices. Discuss its benefits and challenges in hybrid and electric vehicles.	CO3	6
Q. 6 Solve Any Two of the following.			12
A)	Explain the process of matching the electric machine and the internal combustion engine (ICE) in hybrid vehicles. Why is it important?	CO3	6
B)	Describe the role of in-vehicle networks such as CAN (Controller Area Network) in hybrid and electric vehicles.	CO3	6
C)	Discuss the energy management strategies used in hybrid and electric vehicles.	CO3	6